TCP-Based Multi-Client LAN Messaging: A Survey of Transport, Framing, Concurrency and Security, and LANChat-H, an Implemented and Measured Dependency-Light Reference Design

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***Abstract*—A local-area chat application looks trivial from the user's side, yet it forces a designer to solve several networking and security problems at once: recovering message boundaries from a byte stream, serving many clients concurrently, distributing each message to a group, surviving abnormal disconnections, and deciding whom to trust. The literature that addresses these problems is large but fragmented: transport standards describe the byte stream, server-architecture research compares threads with events, messaging standards such as IRC, XMPP, MQTT and WebSocket each fix one design point, and work on reliable groups and secure messaging studies guarantees far stronger than a small system needs. This paper surveys that literature through the lens of a concrete artifact, LANChat-H, a chat system whose core uses only the Python standard library. One TCP server runs a handler thread and a bounded outbound queue with a writer thread per client; messages travel in length-prefixed frames carrying either plain text or versioned JSON events; connections are protected by TLS with trust-on-first-use certificate pinning, memory-hard password hashing, role checks, login lockout and rate limits; history is stored in SQLite; files move over separate, ticket-authorised connections and are verified by SHA-256; and administrative actions are recorded in a hash-chained audit log. On a single laptop over loopback, median delivery latency stayed between 2.7 and 5.8 ms for up to 100 receivers; when one receiver stopped reading, a sequential-relay baseline stopped delivering to all others after about 232 messages whereas the bounded-queue server delivered all 6000 of 6000; private messages, files and, when an administrator switches it on, the group room are end-to-end encrypted with audited primitives (X25519, Ed25519, ChaCha20-Poly1305), so that the server stores only ciphertext, which a test confirms by scanning every frame the server handles and its database, and encrypting a message costs about 0.3 ms; and the code base is covered by 452 automated tests and 370 scripted checks that drive the real window and the terminal client. Forward secrecy, protection of metadata and an asynchronous server variant remain future work.**

***Index Terms*—TCP, socket programming, LAN chat, message framing, length prefix, concurrent server, thread-per-client, head-of-line blocking, TLS, trust on first use, audit log, end-to-end encryption**

I. Introduction

A chat program rarely needs to do anything clever to be useful. Two people type, and each sees what the other typed. Underneath that simple behaviour, however, sits a stack of decisions that every networked application has to make. The server must accept connections and remember who is connected. It must turn a continuous stream of bytes back into the individual messages that were sent. It must handle many clients at the same time without letting shared state become corrupted. It must deliver each message to everyone else in the room, and it must notice and clean up after a client that vanishes without saying goodbye.

TCP is the natural transport, because it offers a reliable, ordered, full-duplex byte stream [2], [1]. The important word is *byte*. The current TCP specification deliberately exposes a stream rather than a sequence of records, so a single `send()` on one side does not correspond to a single `recv()` on the other [2], [7]. A message can arrive split across several reads, and several messages can arrive glued together in one read. Any chat protocol built directly on sockets therefore has to define its own message boundaries.

Several research and standards communities have worked on parts of this problem, each from a different angle. One community specifies the transport itself: reliability, flow control and congestion control [2], [4], [5]. A second community compares ways of serving many connections, from threads and processes to event loops and staged pipelines [11], [12], [14], [13], [17]. A third community defines complete messaging protocols, among them IRC, XMPP, MQTT and WebSocket [18], [20], [23], [22]. A fourth studies what it takes to give a group of processes consistent delivery in the presence of failures [26], [27], [25]. Finally, work on secure messaging explains what an eavesdropper or an impersonator can do to a plaintext channel and what the remedies cost [28], [29], [31].

What stands out when these bodies of work are placed side by side is how little they meet. A protocol standard

tells a reader what a frame looks like, but not how a small server should schedule its handlers. A server-architecture paper compares throughput, but usually on web workloads rather than on a long-lived, many-to-many chat session. A group-communication paper reasons about failure models that a teaching system never approaches. Textbooks do walk through sockets and threads [7], [8], [9], but they present them as separate exercises, and a reader who wants to see framing, concurrency, broadcast and shutdown interacting in one short program has to assemble that picture alone.

This is the gap that motivates the paper. Rather than choosing one of these threads to specialise in, we survey them together, anchored to one concrete artifact: a Python chat system whose core is built with nothing beyond the standard library [32], [33], which we use as a running case. The system began as a minimal threaded relay and was hardened in stages; the version described here is called LANChat-H (H for hardened). The contributions are:

1) a thematic survey of the transport, framing, concurrency, messaging-protocol, reliability, trust and security literature relevant to a small TCP chat system;
2) a comparative analysis of the reviewed approaches and the reference implementation, together with the design decisions they represent;
3) a description of LANChat-H stated precisely from its source code, including what it enforces and what it deliberately omits;
4) an explicit research gap, and the staged hardening path by which the implementation addresses it; and
5) a reproducible evaluation: protocol and security tests, fan-out latency, a head-of-line-blocking experiment against a sequential-relay baseline, file-transfer speed and per-operation security costs, with the code, tests and scripts published online [53] and the limits of the measurements stated.

Section II gives the technical background. Section III reviews and compares the literature. Section IV states the research gap. Section V describes LANChat-H. Section VI presents the evaluation, Section VII discusses limits and future work, and Section VIII concludes.

II. Background

A. TCP as a Byte Stream

TCP establishes connection state before any data moves, numbers every byte so that lost segments can be retransmitted and out-of-order segments reassembled, and uses a window to stop a fast sender from overwhelming a slow receiver [1], [2]. Congestion control, introduced by Jacobson and later standardised, limits how quickly a sender injects data into the network through slow start, congestion avoidance, fast retransmit and fast recovery [4], [5]. Together these mechanisms give an application an ordered, loss-free byte stream.

They do not give it more than that. TCP does not know about users, rooms or messages, it does not persist anything across a process crash, and it does not confirm that a particular chat participant actually read a particular line. This is an instance of the end-to-end argument: functions that only the application can implement completely, such as message boundaries or delivery confirmation to a person, cannot be delegated wholesale to the lower layer [6].

TABLE I
Application-Layer Framing Alternatives

Method	Advantage	Limitation
Fixed size	Trivial parsing	Wasted space; inflexible
Delimiter	Human-readable; simple for lines	Escaping and delimiter collision
Length prefix	Binary-safe; exact; variable size	Length must be validated
TLV	Extensible typed fields	Larger parser and schema
Self-describing	Rich structure (XML/JSON)	Encoding overhead; needs outer framing

B. Connection Failure and Liveness

A TCP connection can end cleanly, with an exchange of FIN segments, or it can fail through a reset, a timeout, a crashed process, a dropped interface or a network partition. An application normally learns about failure when a read returns end-of-stream, when a send or receive raises an exception, or when a liveness policy expires. TCP keep-alive is optional and off by default; the host-requirements standard requires that it be configurable and warns that a single unanswered probe is not proof that the peer is gone [3]. In practice this means that explicit quit commands and exception handling handle the common case, whereas noticing an unplugged laptop quickly needs application-level timeouts or heartbeat messages.

C. Why Framing Is Mandatory

Because TCP segmentation is independent of application writes, a receiver cannot rely on read boundaries. Four framing strategies dominate practice, summarised in Table I. *Fixed-size* records are trivial to parse but waste space and cannot carry variable payloads. *Delimiter*-based framing, used by line-oriented protocols such as IRC, is readable but needs an escaping rule whenever the delimiter can occur inside the data [18]. *Length-prefixed* framing announces the payload size before the payload, so a receiver reads exactly that many bytes; it is binary-safe and precise, at the price of having to validate the announced length. *Self-describing* formats, such as the XML streams of XMPP, carry structure inside the data but still depend on an outer mechanism to find where one unit ends [20]. WebSocket and MQTT both use length information in binary headers [22], [23].

For a stream of UTF-8 text [10], a four-byte unsigned length in network byte order followed by the encoded payload is compact and easy to test. Python's `struct` module defines the `!` format prefix as network (big-endian) order with standard sizes, so `"!I"` is an interoperable four-byte header [34]. The receiver must loop until the requested number of bytes has accumulated or the peer closes, because a single `recv()` may return fewer bytes than asked for. Fig. 1 shows the idea, and Listing 1 shows the code used by the reference implementation.

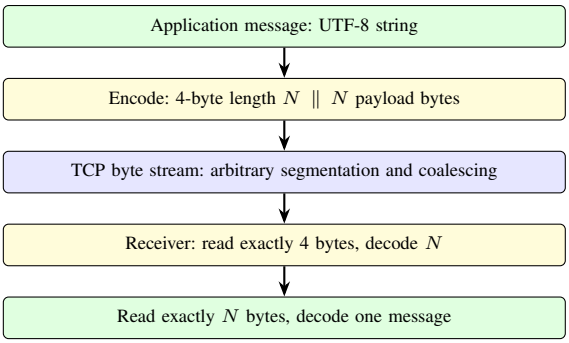


Fig. 1. Length-prefixed framing over a TCP byte stream.

```
MAX_MESSAGE_BYTES = 4096

def send_message(conn, message):
    payload = message.encode("utf-8")
    if len(payload) > MAX_MESSAGE_BYTES:
        raise ValueError("message is too large")
    conn.sendall(struct.pack("!I", len(payload)) + payload)

def receive_message(conn):
    header = _receive_exactly(conn, 4)
    if header is None:
        return None             # peer closed
    (length,) = struct.unpack("!I", header)
    if length == 0 or length > MAX_MESSAGE_BYTES:
        raise ValueError("invalid message length")
    payload = _receive_exactly(conn, length)
    if payload is None:
        return None
    return payload.decode("utf-8")
```

Listing 1. Framing logic of the reference implementation (`protocol.py`, condensed).

D. Concurrency Models for Servers

A chat server holds many connections open for a long time, and most of the time each one is idle. Four models are usually contrasted. *Thread-per-client* gives each connection a sequential control flow that is easy to read and debug, at the cost of one stack and one scheduling entity per client. A *thread pool* bounds the number of workers but adds queueing and makes long-lived connections awkward. An *event loop* multiplexes many connections on few threads using readiness notification, which scales further but forces the programmer to express each connection as a state machine [17], [16]. *Staged* architectures split a service into event-driven stages joined by explicit queues with dynamic resource control [12]. The debate between threads and events is old and unresolved: Ousterhout argued that threads are the wrong abstraction for most purposes, von Behren *et al.* argued that the objections are really objections to particular thread implementations, and Lee described the difficulty of reasoning about shared-memory threads [13], [14], [15].

E. Group Delivery and Topology

Delivering one message to many recipients can be done in several ways. A central server can relay by repeated unicast, sending a separate copy on each connection. Peer-to-peer designs remove the relay but introduce membership, connection-count and consistency problems. IP multicast sends one datagram to a group, but it uses datagram semantics, which moves loss recovery, ordering and membership back to the application [24]. Broker-based publish–subscribe decouples senders and receivers in space, time and synchronisation [25]. Repeated unicast is not Ethernet broadcast or IP multicast: every recipient has an independent TCP connection with its own flow-control window and failure state.

F. Security Considerations

A trusted LAN reduces deployment effort but is not a security mechanism. On a plaintext channel, any reachable host can connect, choose any name, inject messages, and read traffic. TLS 1.3 protects a client–server channel against eavesdropping, tampering and forgery and can authenticate the server with a certificate [28]. It does not authenticate individual users, and the server still sees plaintext after decryption. End-to-end secrecy between users needs a different construction, such as the Double Ratchet used by Signal, which provides forward secrecy and post-compromise security [31], [30]. Unger *et al.* give a systematic comparison of secure-messaging designs [29].

Three further mechanisms matter for a small multi-user server. First, trust in a self-signed server certificate can be established on first use and then pinned, the model that SSH uses for host keys [38]; it protects every later connection but not the first. Second, passwords should be stored only as salted, memory-hard hashes such as scrypt [41], and online guessing should be throttled, as recommended for authentication systems generally [48]. Third, an audit log whose entries are chained by hashes makes later alteration detectable [49], [50].

III. Literature Review

Rather than reviewing sources one by one, it is more useful to group them by the design question they answer. Table II compares the approaches discussed below, and the reference implementation, on the properties most relevant to a small LAN chat system.

Complete messaging protocols. IRC was the first widely deployed text chat protocol on TCP. Its base specification frames messages as lines ending in carriage return and line feed, limited to 512 characters, and connects servers in a network so that users on different servers can talk to each other [18], [19]. The design is easy to implement by hand, which explains its longevity, but the delimiter approach forces every reader to agree on how to handle over-long or malformed lines. XMPP takes the opposite direction: it defines a long-lived XML stream over TCP with negotiation of TLS and SASL authentication, and a separate specification adds rosters and presence for instant messaging [20], [21]. That richness is precisely what makes it heavy for a small system. MQTT 5.0 places a broker between publishers and subscribers and adds quality-of-service levels, sessions and a compact binary header [23]. WebSocket adds an HTTP-compatible opening handshake and then exchanges length-carrying frames, which makes it the natural route to browser clients [22]. All

TABLE II
COMPARATIVE ANALYSIS OF REVIEWED APPROACHES

Approach	Transport	Framing	Topology	Security	Persistence	Key limitation for a small teaching system
IRC [18], [19]	TCP	CRLF lines, 512-byte limit	Server network	None in base spec	None	Text-only framing; limits fixed by the spec
XMPP [20], [21]	TCP	XML stream	Federated client–server	SASL and TLS	Via extensions	Heavy parser; large specification surface
MQTT 5.0 [23]	TCP (optionally TLS)	Binary header, variable length	Broker pub/sub	Optional; credentials in CONNECT	Sessions and QoS	Needs a broker; more machinery than a chat room needs
WebSocket [22]	TCP after HTTP upgrade	Length-carrying frames; client masking	Client–server	TLS via `wss`	None	Needs an HTTP handshake and a framework
Signal / Double Ratchet [31], [30]	Any	Not specified	Relayed	End-to-end	Server-dependent	Cryptographic design, not a transport or server design
FTP [37]	TCP, separate control and data connections	Commands as text lines; data as a raw stream	Client–server	None in base spec	Files only	Predates TLS; no chat, presence or history
Flash [11]	TCP (web)	HTTP	Client–server	N/A	N/A	Web workload; compares process, thread and event models
SEDA [12]	TCP (services)	Application-defined	Staged pipeline	N/A	N/A	Targets far larger loads than a classroom chat
Threads vs. events [13], [14], [16]	TCP	N/A	N/A	N/A	N/A	Position papers, not chat-specific
Reliable groups [26], [27]	Reliable channels	N/A	Process groups	N/A	Varies	Guarantees far beyond a demonstration system
LANChat-H (this work)	**TCP + TLS**	**4-byte length + text or JSON events**	**Central relay**	**TLS with pinning, scrypt, roles, audit, end-to-end encryption**	**SQLite**	**Single server; no forward secrecy; metadata visible to the server (Sec. VII)**

four solve interoperability and deployment problems that a command-line demonstration deliberately sidesteps.

Server-architecture research. Pai, Druschel and Zwaenepoel compared multi-process, multi-threaded, single-process event-driven and asymmetric event-driven designs in the Flash web server and showed that architecture materially changes throughput and blocking behaviour [11]. SEDA proposed decomposing a service into stages linked by bounded queues, with dynamic controllers that keep the system stable under overload [12]. Position papers on both sides of the threads-versus-events debate followed [13], [14], [15], and Krohn *et al.* argued that event-driven code can be written with a structure close to threaded code [16]. The C10K discussion collected the operating-system techniques needed to serve ten thousand simultaneous connections [17]. What these works share is that their target workloads are much larger and more heterogeneous than a room of a few terminals. They justify an architectural boundary, meaning the point at which thread-per-client stops being adequate, without proving that threads are the wrong choice below it.

Group communication and reliability. Birman and Joseph's work on virtual synchrony and the later process-group approach show what *reliable* group delivery means when processes can fail: members agree on group views and on the order in which messages are delivered relative to membership changes [26], [27]. Eugster *et al.* survey publish–subscribe and classify it by how it decouples producers and consumers [25]. A chat server that relays by repeated unicast provides none of these guarantees. A message accepted by the server can reach some clients before another connection fails, and there is no acknowledgement set, replay or globally stable order. Distinguishing this from transport reliability is important, because TCP's reliability is easily mistaken for delivery reliability.

Secure messaging. TLS 1.3 secures the channel [28], and the secure-messaging literature separates transport security, user authentication, authorisation, and end-to-end secrecy as distinct layers [29]. Signal's Double Ratchet shows how end-to-end secrecy with forward secrecy can be combined with an untrusted relay [31], [30]. These layers are additive, and a system can adopt them in stages. The same layering applies to groups: the X3DH key agreement gives asynchronous contact set-up between parties that need not be online together [47], and the Messaging Layer Security protocol addresses efficient group key management with member addition and removal [46]. Both rest on the same primitives that are available in widely reviewed libraries: X25519 key agreement [42], Ed25519 signatures [43], HKDF [45] and ChaCha20-Poly1305 [44].

Credentials, trust establishment and audit. Small systems

usually treat accounts as an afterthought. Standards guidance on authentication recommends salted, memory-hard password hashing, a minimum length with a check against common passwords, and limits on repeated failed attempts [48], [41]. For servers with self-signed certificates, SSH's trust-on-first-use host-key model is the most widely deployed precedent [38]. For accountability, tamper-evident logging work shows that chaining each entry's hash to its predecessor lets an auditor detect modification, deletion or insertion, with a separate commitment to the newest entry needed to detect truncation [49], [50]. These ideas appear in operating systems and services, but rarely in a teaching-scale chat server where all of them can be read together.

Discovery and bulk transfer. On a LAN, services are normally found with multicast DNS and DNS-based service discovery [39], [40]; small programs often use a custom UDP broadcast instead, which is simpler but limited to one broadcast domain. For files, FTP established the pattern of a control connection for commands and a separate data connection for bytes [37], so that a large transfer never blocks or overflows the command channel. Modern chat protocols that carry media over their message channel must either fragment it or rely on a side channel for the same reason.

Foundational texts. Stevens' treatment of the sockets API and the standard networking textbooks remain the main sources for how a program actually uses TCP [7], [8], [9].

Read together, these groups trace a fairly clean progression: from byte-stream transport, to explicit message framing, to concurrent servers, to full messaging protocols, and finally to reliable and secure group communication. What none of them does individually is present a compact, dependency-free system in which all of these decisions interact and can be inspected and measured at once.

IV. Research Gap

Reading Table II left to right, every reviewed approach is strong in the columns it was designed for and silent elsewhere. The remainder of this section explains why closing the gaps is a matter of design, not of adding one more feature.

Framing, concurrency and shutdown are studied separately. Protocol standards specify frames, architecture papers compare schedulers, and textbooks teach sockets and threads as independent exercises. In a real chat server they interact: a handler thread blocks inside a read while another thread wants to shut the server down, a broadcast iterates a client table while another handler removes an entry, and a half-received frame is abandoned when a peer disconnects mid-message. Few sources examine these interactions in a system small enough to read in one sitting.

Sequential broadcast has a head-of-line risk. When a handler sends a message to every recipient in turn, a slow receiver whose send buffer is full can delay delivery to every recipient after it. Per-client outbound queues with writer workers, bounded queues with a disconnect-or-drop policy, or non-blocking writes isolate slow consumers. This is well known in large systems [12], but it is rarely shown in a teaching-scale implementation together with the code and tests that demonstrate the problem.

Liveness detection is left to defaults. Because TCP keepalive is optional and off by default [3], a suspended laptop or an unplugged cable can leave a connection half-open for a long time. Explicit exit and exception cleanup cover normal termination, but abnormal termination needs a heartbeat or timeout that is seldom included in small examples.

Security is either absent or arrives all at once. Simple demonstrations are plaintext with no identity; production systems bring TLS, accounts and end-to-end encryption together [28], [29], [31]. The intermediate steps, such as wrapping sockets with the standard `ssl` module and adding a name-uniqueness rule, are rarely presented as a staged path with a clear statement of what each stage does and does not protect.

Evaluation is anecdotal. Very few small messaging examples come with a protocol-correctness test that fragments and coalesces frames, or with a measurement method for latency and fan-out cost. Without one, no claim about when thread-per-client stops being adequate can be checked.

LANChat-H (Section V) is built around the premise that these gaps share one root cause: transport, framing, concurrency, delivery and security are presented as separate problems, when in a chat system they are stages of one path that must agree on message size limits, lifecycle and failure handling. Section VI reports what happens when that path is built and measured.

V. LANChat-H: The Reference Implementation

LANChat-H is the case-study system. It began as a minimal command-line relay (a TCP server, one thread per client, a four-byte length prefix and plaintext broadcast, with three tests) and was extended in stages, each independently testable. This section states what the current source code does; Section VI measures it. The core uses only the Python standard library [32], [33], [34], [35]. The standard-library modules `sqlite3` and `hashlib` supply storage and scrypt [51], [52], and `tkinter` supplies the desktop window. Three third-party packages are optional: `cryptography` for end-to-end encryption, Pillow for some picture previews and `sounddevice` for voice recording.

A. Architecture

Fig. 2 shows the structure. The server owns membership, persistence and policy; clients are thin. In total the non-graphical code is about 4,800 lines in 21 modules, the window about 4,900 lines in seven modules, and the end-to-end encryption code about 900 lines in three modules. Table III maps modules to the concepts surveyed earlier.

B. Framing, Protocol Versions and Compatibility

Every message in both directions is a four-byte big-endian length followed by UTF-8 text, as in Listing 1; zero-length and over-long frames close the connection. Two protocol versions share this framing. Version 1 carries plain text lines and is what the terminal client speaks. Version 2 carries

TABLE III
IMPLEMENTATION MAPPING AND HONEST SCOPE

Component	Concept	Role in the system	Boundary / not provided
`protocol.py`, `events.py`	Record recovery; typed events	4-byte network-order length, exact-read loop, 4096-byte limit; JSON events with a type field, plus a renderer that turns any event into a plain line for old clients	No acknowledgements or sequence numbers on the wire
`server.py`, `server_admin.py`, `server_keys.py`, `server_room.py`	Accept, membership, fan-out, policy	Handler thread and bounded outbox with writer thread per client; handshake; history, search, pins; ticketed file transfer; admin and key-directory events	Single server; no failover; one thread per client
`store.py`, `accounts.py`, `permissions.py`	Persistence, credentials, roles	SQLite with six numbered migrations, FTS5 search with `LIKE` fallback; scrypt hashes; admin, member and guest roles checked on the server	One connection behind a lock; not a clustered database
`security.py`, `audit.py`	Abuse resistance; accountability	Lockout with backoff, token-bucket rate limits, connection and sign-up caps, password rules; hash-chained audit log	In-memory counters reset on restart
`storage.py`, `transfer.py`	Bulk transfer	Content-addressed blob store, name sanitising, blocked types, streaming SHA-256 verification	Files are not encrypted at rest on the host
`chat_core.py`, `certs.py`, `discovery.py`	Client handshake, TLS trust, discovery	Login, certificate pinning by SHA-256 fingerprint, self-signed certificate creation, UDP broadcast discovery	Discovery works within one broadcast domain
`e2ee.py`, `e2ee_group.py`, `keystore.py`	End-to-end encryption	Identities, passphrase-wrapped key file, recovery code, safety numbers, pinned contacts, sealed messages and files, the room's key chains	No forward secrecy; the server still sees metadata (Sec. VII)
`gui_*.py`, `client.py`, `chat.py`, `admin_cli.py`	Interface	Desktop window, terminal client, single launcher, offline admin tool	Terminal client lacks edits, reactions, search and pictures

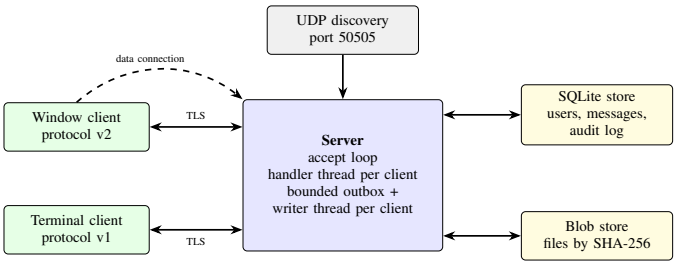


Fig. 2. Architecture of LANChat-H. Both client kinds hold one TLS connection to a central relay; file bytes use a second, short-lived data connection authorised by a ticket from the first.

one JSON object per frame with a type field (message, edit, delete, reaction, typing, presence, read marker, pin, search, history page, upload, download and so on); a version-2 client announces itself with a JSON hello in the username step, and the server detects the version from the first character. For version-1 clients the server renders every event as the line they have always seen, and returns nothing for events a terminal cannot show, so new features were added without breaking the old client. Message text is limited to 3400 bytes so that the JSON envelope always fits one frame.

C. Concurrency and Delivery

Each connection has a handler thread that reads requests and a writer thread that is the only code writing to that socket. Delivering a message means placing it on every recipient's bounded outbound queue (default 100 entries) without waiting. If a queue fills, the client is not keeping up and is disconnected; it can reconnect and replay history from the database. This is the Stage 1 isolation of slow receivers described in the original design, and Section VI measures its effect. The welcome and history are written directly before the writer thread starts, so they arrive first and in order. Clients send a ping every ten seconds and the server drops a connection silent for thirty. A user name may be signed in once; a saved sign-in may take over a stale connection.

D. Security Design

Table IV lists each threat with its control; the controls are in code and covered by tests (Section VI).

E. Persistence and Features

Messages, users, reactions, pins, read markers, sessions and the audit log live in SQLite, so history survives restarts; six numbered migrations let an older database upgrade in place. A joining client receives the last 50 messages and can page back 50 at a time. Search uses an FTS5 index and falls back to `LIKE` where FTS5 is absent. The window supports replies, reactions, edit and delete (deletion keeps a tombstone so quoted replies still resolve), typing indicators, presence, read receipts, mentions, pins and an administrator console. These features matter to the survey chiefly because each stresses the protocol: history paging and search require back-pressure-aware replies, and edits and deletes require ids that survive reconnection.

F. Bulk Transfer

Pictures, documents and audio do not travel on the chat connection. The client sends an upload request with the name,

TABLE IV
THREATS AND CONTROLS IN LANCHAT-H

Threat	Control
Reading or altering traffic	TLS (1.2 or newer) on chat and data connections. The host creates a self-signed certificate once.
Impersonating the server	Trust on first use: the client shows the certificate's SHA-256 fingerprint, pins the certificate after approval and refuses a different one later. Host name checking is off because LAN addresses change; the pinned certificate authenticates the server.
Stolen account database	Passwords stored only as salted scrypt hashes ($n = 2^{14}$, $r = 8$, $p = 1$, 32-byte output) with constant-time comparison.
Password guessing	Five wrong passwords lock a name from an address for 30 s, doubling to 15 min; one address failing for many names is locked after 20; a correct password clears the count. New passwords need six characters, must differ from the user name and common passwords; at most ten sign-ups per address per hour.
Flooding	Per-person token bucket (30 burst, 10 per second); 240 new connections per address per minute; frame, message and file size limits.
Privilege abuse	One permission table; every action checked on the server; the last admin cannot be demoted or disabled.
Password kept in memory	After sign-in the window keeps a random token (12 h), stored on the server only as a SHA-256 hash, revoked on leave, password change, disable or reset.
Hostile files	Names reduced to a base name; files stored under their SHA-256; executable types blocked; size and space limits enforced while streaming; hash checked on upload and on download; tickets are single-use, expire in 60 s and are limited per person.
Disputed history	Audit log of sign-ins, failures, lockouts, role changes, deletions of others' messages, uploads, resets and settings, each row hashed with its predecessor; a head marker detects truncation; message text is never logged.
The host reading chats and files	End-to-end encryption of private messages, files and (optionally) the group room: the server stores ciphertext and public keys only (Sec. V-G).
A fake key from the server	Self-signed key bundles, pinning on first sight, sending blocked after a change until accepted, sixty-digit safety numbers.
Replays and forged senders	Every encrypted message is signed and bound to its sender and recipients (or epoch and number); the window flags reused ids and numbers.
A removed member	A new key epoch for the room, whose keys the removed person never receives.
An unattended window	Auto-lock after an idle time covers the window and forgets the keys.

size and SHA-256; the server checks role, type, size, quota and conversation and replies with a one-time ticket. The client opens a second TLS connection, presents the ticket, and streams raw bytes; the server writes them to a temporary file while hashing, enforces the declared size and a deadline, compares the digest, and only then moves the file into place under its hash and posts the message. Downloads mirror this and are verified by the receiver; a damaged file is never delivered, and a half-finished upload leaves nothing behind. Identical content is stored once, and a file is removed when the last message referring to it is deleted.

G. End-to-End Encryption

Each person has an X25519 key pair for key agreement and an Ed25519 pair for signing, bound by a self-signature [42], [43]. Secret keys are wrapped on disk with ChaCha20-Poly1305 [44] under a key derived by scrypt from a passphrase. A 24-character recovery code (about 119 bits) wraps a second copy that the server stores as an opaque backup; because the code is high entropy the server cannot usefully guess it, and a test confirms that neither the keys nor the code appear in the database file. The server keeps only public keys and cannot be trusted to report them honestly, so clients verify each bundle's signature, pin the first key seen for each contact, block sending when it changes until the person has accepted the new one, and offer a sixty-digit safety number computed from both identities.

Private messages. The sender encrypts the text once under a random key with ChaCha20-Poly1305, seals that key separately for the recipient and for themselves (ephemeral and static X25519 through HKDF-SHA256 [45]), and signs the whole message with Ed25519, binding the sender and recipients into the associated data. The server checks only the shape and the two names, stores an empty body and the envelope, and relays it. Tampering, a wrong recipient or a forged sender fails to open, and the window flags a reused random message id as a possible replay. **Files** are encrypted on the sender's computer in 64 KiB chunks under a random key; each nonce carries the chunk number and a last-chunk flag, so dropped, reordered or added chunks fail. The key, the real name and the type travel inside the encrypted message, so the server stores unreadable bytes under an opaque name, and the receiving window, which alone knows the name, warns before opening a program-like file.

The group room. When an administrator turns it on, the room uses sender keys in the style of group messengers [46]: each member holds a random chain key per epoch, derives each message key by a one-way step, and gives every other member a sealed copy of the chain as it stands, so a member who joins later reads only what follows. The server stores the sealed copies, enforces the current epoch and increasing message numbers, and starts a new epoch whenever someone is removed or an administrator asks, so the removed person never receives the new keys. Mentions are named by the sender because the server cannot read the text; pins, quoted replies and search use what the window has decrypted.

Key lifecycle. Keys can be rotated (a new identity, with the old ones kept so earlier messages still open) or restored from the recovery code on a new computer, and an idle window locks itself, covering the interface and forgetting the keys until the passphrase is entered.

TABLE V
HARDENING STAGES AND THEIR STATUS

Stage	Status in LANChat-H
1. Robustness	Done: bounded per-client queue with writer thread, heartbeat and idle timeout, unique names, fragmentation and disconnect tests
2. Protocol structure	Done: typed JSON events, version negotiation, text compatibility for old clients
3. Security and identity	Done: TLS with pinning, scrypt credentials, roles, lockout, rate limits, audit log
4. Asynchronous variant	Not done; listed as future work
Beyond the plan	Persistence, file transfer, administration, end-to-end encryption

H. Status of the Staged Hardening Path

The system was hardened in the order shown in Table V. The only stage not carried out is the asynchronous variant, so the threaded design has not been compared with an `asyncio` server.

VI. EVALUATION

All measurements were taken on one machine: an Apple-silicon laptop (arm64) running macOS 27.0.1 with CPython 3.13.9, with the server and clients on the loopback interface. The source code, tests and scripts (`bench/`) are published [53], and the numbers below are what they printed. Because client and server share one machine, the results show the cost of the software, not of a real network; they are not claims about Wi-Fi or Ethernet performance.

A. Correctness and Security Tests

The repository holds 22 test modules that run 452 tests, each starting real servers on free local ports. They cover framing (round trip, closed connection, oversized and zero-length frames), the version-1 and version-2 protocols, the database and its migrations, search and paging, TLS and certificate pinning including a changed certificate, discovery, file transfer (round trips, hash mismatch, cut-off uploads, hostile names, expired and reused tickets, quota, concurrent uploads), lockout, rate limits, password rules, saved sign-ins, administrator actions, tamper detection in the audit chain, and the encryption code, which is checked against published test vectors for X25519 and Ed25519 [42], [43] and then against tampering, wrong recipients, forged senders, replays, damaged and reordered files, and the room's rules for late joiners, removal and message numbering. One test wraps the server's send and receive functions, records every frame the server handles, and scans those frames, the database file and the stored files for the secret text, file names, file contents and file keys of encrypted messages; a companion check shows that the same capture would reveal a plaintext message. A further eleven scripts drive the real window and the terminal client through 370 checks, including runs with every security default on. All of them pass.

TABLE VI
BROADCAST LATENCY VS. RECEIVERS (LOOPBACK, MS)

N	TLS	**Median**	**p95**	**Max**	Delivered
1	no	3.23	4.64	6.98	100 %
5	no	3.67	4.93	16.65	100 %
10	no	2.71	5.23	7.78	100 %
25	no	2.72	5.91	11.51	100 %
50	no	4.16	6.65	26.70	100 %
100	no	5.83	9.40	13.94	100 %
1	yes	2.88	4.69	6.00	100 %
25	yes	4.66	7.46	13.59	100 %
100	yes	6.44	11.32	21.33	100 %

B. Fan-Out Latency

One sender sent 100 messages of about 100 bytes at ten per second to N receivers, each a real client with its own connection; every message was also written to the SQLite database before delivery. Receivers ran in separate processes of at most 20, and latency is the arrival time minus the send time on the machine's monotonic clock. Table VI shows that every message reached every receiver. Median latency stayed between 2.7 and 5.8 ms and the 95th percentile below 9.5 ms without TLS; with TLS the figures for 25 and 100 receivers were 4.7 and 6.4 ms (median) and 7.5 and 11.3 ms (95th percentile).

C. Head-of-Line Blocking

The fan-out cost model predicts the effect of a stalled receiver. With sequential sends, recipient k receives the message after

$$D_k = \sum_{j=1}^{k} s_j, \qquad T_{\text{bcast}} = D_N, \tag{1}$$

where s_j is the time the blocking send to recipient j takes, and if recipient m stalls for an extra time W then

$$D_k \;\geq\; W + \sum_{j=1}^{k} s_j \qquad \text{for all } k \geq m. \tag{2}$$

With a bounded queue and writer thread the send to each recipient reduces to an enqueue with cost $q_j \ll W$, so D_k no longer depends on W for $k \neq m$.

To test this we compared LANChat-H with a minimal sequential relay written for the experiment: a thread per client that sends each incoming message to every other client in turn with a blocking `sendall`. Both use the same framing. Ten receivers read normally; one more connects first, shrinks its receive buffer to 4 KiB and never reads. A sender sends 600 messages of 2000 bytes at 50 per second. Table VII gives the range over four runs. Once the kernel buffers for the stalled connection were full, after 232 messages (about 460 kB), the sequential relay stopped delivering to every receiver: exactly 2320 deliveries (232 messages to each of ten receivers) arrived in every run, out of 5370 to 5400 expected, and the sender's own sends began to time out after 537 to 540 messages. With the stalled receiver present LANChat-H delivered 6000 of

TABLE VII
ONE STALLED RECEIVER, TEN HEALTHY ONES

Server	Stalled	Delivered	Median ms	p95 ms
Sequential relay	no	6000/6000	0.65–0.78	1.36–5.46
Sequential relay	yes	2320/5370–5400	0.61–0.85	1.38–13.03
LANChat-H	no	6000/6000	3.75–4.36	5.36–8.82
LANChat-H	yes	6000/6000	3.16–4.58	5.53–5.92

TABLE VIII
FILE TRANSFER THROUGHPUT (LOOPBACK, MB/S, MEDIAN OF 3)

File (MB)	TLS	Upload	Download
1	no	282	636
1	yes	171	303
5	no	409	539
5	yes	370	464
20	no	616	595
20	yes	394	461

6000 messages to the ten healthy receivers, and dropped the stalled connection once its queue filled.

The table also shows the price. In the absence of a stall the sequential relay is faster per message (median under 1 ms against 4 ms), because it does no more than forward bytes, whereas LANChat-H also persists each message, checks the sender's role and rate limit, and renders it for each protocol version. Isolation of slow receivers is therefore a trade of a few milliseconds of latency for the guarantee that one bad connection cannot stop the room.

D. File Transfer Speed

Table VIII gives the median of three runs for uploading and then downloading a random file through the ticket and data-connection path, including hashing, the ticket exchange and verification. Throughput is in the hundreds of megabytes per second on loopback and TLS costs roughly a third for small files and less for large ones. These figures show that the design adds no obvious bottleneck; they say nothing about a wireless link, which would dominate in practice.

E. Cost of Security Operations

Table IX lists the median time of single operations. A password check costs about 27 ms by design: it is the price of making offline guessing expensive. Encrypting a message for one person (a random key sealed twice and a signature) takes about 0.3 ms and decrypting it 0.26 ms, a room message under 0.3 ms, and a 20 MB file is encrypted in about 31 ms and decrypted in about 23 ms (roughly 650 and 850 MB/s). The cost of end-to-end encryption is therefore dominated by passphrase stretching and key management, not by the ciphers, and the server's own work is unchanged: it relays the same frames.

F. Threats to Validity

The experiments use one machine, one Python version and the loopback interface, so queueing in the network, wireless loss and operating-system differences are absent. Latency includes scheduling of separate Python processes and is not a pure network figure. The synthetic load is regular (fixed rate and size), not a trace of real conversations. The sequential relay is a minimal baseline written for the comparison, not a measurement of any other published system, and the claim that thread-per-client stops being adequate at some size has not been tested because no asynchronous variant was built. The tests show that behaviours occur; they do not prove their absence elsewhere.

TABLE IX
COST OF SINGLE OPERATIONS (MEDIAN OF 15 RUNS)

Operation	Time (ms)
Password check (scrypt $n = 2^{14}$)	27.3
Wrap secret keys with a passphrase (scrypt $n = 2^{15}$)	57.7
Generate an identity (X25519 and Ed25519)	0.05
Safety number (5200 SHA-512 rounds)	2.1
Encrypt a 200-byte message for one person	0.31
Decrypt it	0.26
Room message: encrypt / decrypt	0.17 / 0.26
Seal a chain key for one member	0.25
Walk a room chain forward 1000 steps	1.3
Encrypt / decrypt a 20 MB file	30.7 / 23.4

VII. LIMITATIONS AND FUTURE WORK

No forward secrecy. Private messages use an ephemeral key per message, which protects the sender's side, but theft of a recipient's long-term key would expose earlier messages to anyone holding the ciphertext, and the room's sealed chain copies stay on the server so that history can be re-read. A ratchet [31], [30] would be the upgrade, at the cost of storing per-device state.

Identity is as strong as the comparison. Keys are pinned on first sight and a change blocks sending, but until two people compare security codes a hostile server could have substituted a key at the very first contact.

Scope of encryption. Terminal clients and members without keys cannot take part in encrypted conversations; the server can still withhold or delete messages even though it cannot read or forge them; search of encrypted messages covers what the window has loaded; and files shared in readable conversations are stored unencrypted.

Metadata remains visible to the server: who talks to whom, when, message sizes, reactions and mentions.

First-connection trust. Trust on first use fails if an attacker is already present at the first connection and nobody compares the fingerprint.

Scale. The design is one server, one thread per client and one SQLite connection behind a lock. The paper does not establish where this stops being adequate; an `asyncio` server [36] under identical workloads, measured across a real network and with more clients, is the obvious next experiment.

Evidence. The evaluation is single-host. A multi-host study, including packet loss and wireless links, would test the head-of-line and transfer results under realistic conditions.

VIII. Conclusion

Taken together, the literature on TCP-based messaging tells a story of solid but scattered progress. Transport standards define a reliable byte stream, framing conventions recover messages from it, server-architecture research explains when threads give way to events, messaging protocols fix particular design points, and group-communication and security work describe guarantees well beyond a demonstration system. What the comparison and gap analysis point to is that no single small artifact walked through all of these decisions together in a form that could be read, tested and measured.

LANChat-H is an attempt to be that artifact. Its length-prefixed framing, versioned events with a text fallback, bounded per-client queues, heartbeats, certificate pinning, memory-hard credentials, throttling, ticketed file transfer, hash-chained audit log and end-to-end encryption of private chats, files and the group room were each added in a testable stage, and the measurements show the effects the survey predicts: a sequential relay stops delivering to everyone once one receiver stalls, whereas bounded queues keep the room alive at the cost of a few milliseconds per message. The honest remainder is stated as plainly: there is no forward secrecy, the server still sees metadata, no asynchronous variant was compared, and all measurements come from one machine. The next phase is to add a ratchet and to repeat the evaluation across a real network.

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